

Playing Sudoku on random 3-regular graphs

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Introduction

Main Results

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Analysis

Conclusion

Introduction

The “hardest” Sudoku

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The “hardest” Sudoku

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			8		1			
							4	3
5								
				7		8		
						1		
	2			3				
6							7	5
		3	4					
			2			6		

Answer: **17**, proved with computer assistance by McGuire, Tugemann, & Civario (2013).

Sudoku on a graph

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- there exists a partial $\chi(G)$ -colouring $c : S \rightarrow \{1, \dots, \chi(G)\}$;
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The **Sudoku number** of G is

$$s(G) := \min\{|S| : S \text{ is a Sudoku set for } G\}.$$

(Terminology follows (Lau et. al. 2022), but the parameter goes by many names.)

Sudoku on a graph

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- random graphs?

We studied the Sudoku number of the random 3-regular graph $\mathcal{G}_{n,3}$.

Main Results

The Sudoku number of a random 3-regular graph

Theorem (Dippel, E., Prałat, Willhalm)

A.a.s. (with probability tending to 1 as $n \rightarrow \infty$),

$$s(\mathcal{G}_{n,3}) \leq (1 + o(1)) \frac{n}{3}.$$

(Note: $\chi(\mathcal{G}_{n,3}) = 3$ a.a.s.)

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Proof Approach

- Switch to an easier model.
- Design an algorithm which, a.a.s., properly 3-colours $\mathcal{G}_{n,3}$ and simultaneously constructs a corresponding Sudoku set.
- Analyze algorithm using *differential equations method*.

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Any 3-regular graph G with n vertices and $\chi(G) = 3$ has

$$s(G) \geq \left\lceil \frac{n+2}{4} \right\rceil.$$

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Conjecture

$$s(\mathcal{G}_{n,3}) = (1 + o(1)) \frac{n}{4} \quad \text{a.a.s.}$$

- Supported by simulations
- Related to bootstrap percolation/“decycling number”

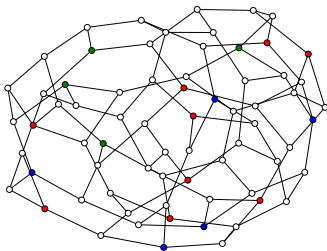
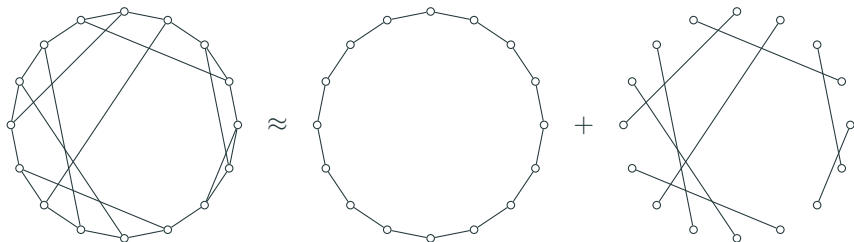


Figure 1: Minimal Sudoku set for $n = 70$ (size $18 = \lceil 70/4 + 1/2 \rceil$)

The algorithm

A simpler model: cycle plus random matching

$$\mathcal{G}_{n,3} \approx \mathcal{H}_n + \mathcal{M}_n$$



An event A_n holds a.a.s. in $\mathcal{G}_{n,3}$ iff it holds a.a.s. in $\mathcal{H}_n + \mathcal{M}_n$.

The “greedy Sudoku set”

Process vertices in the cyclic order of $\mathcal{H}_n$ and partially reveal $\mathcal{M}_n$ sequentially.



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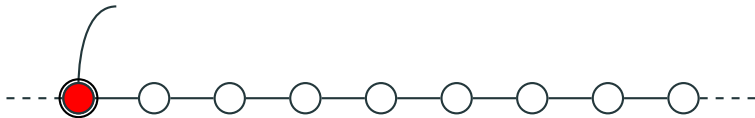
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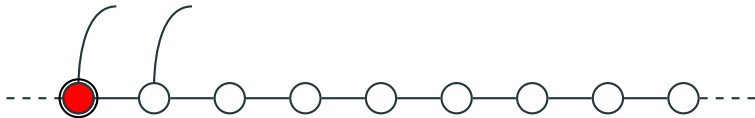
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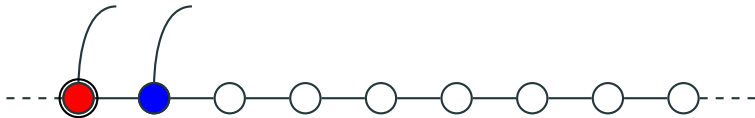
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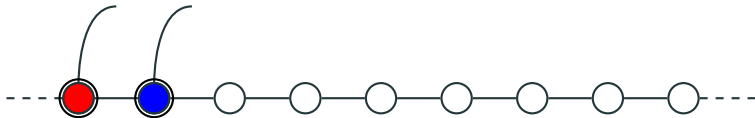
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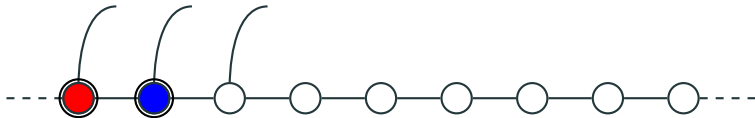
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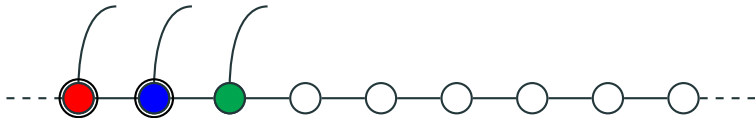
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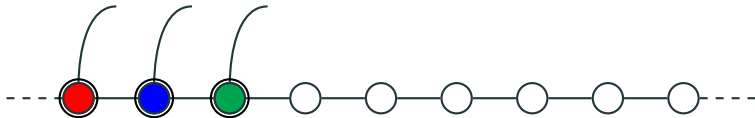
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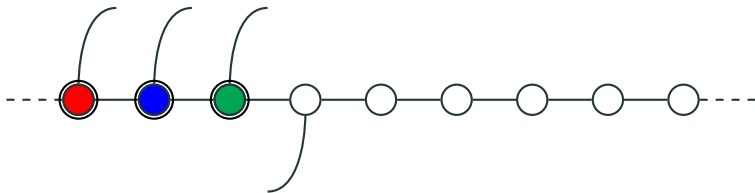
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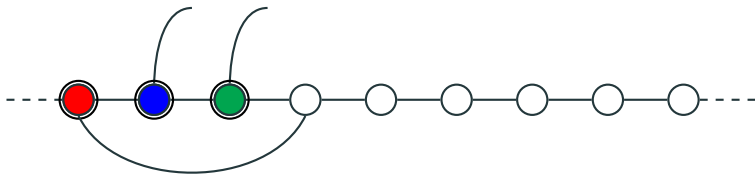
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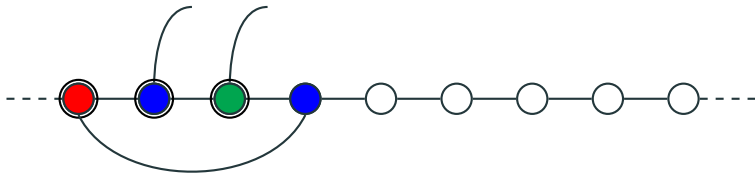
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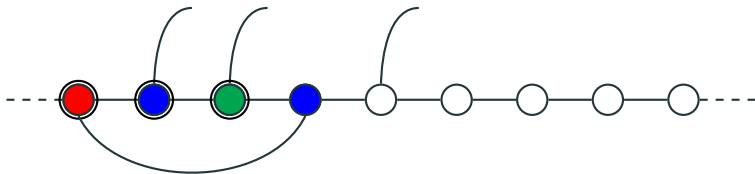
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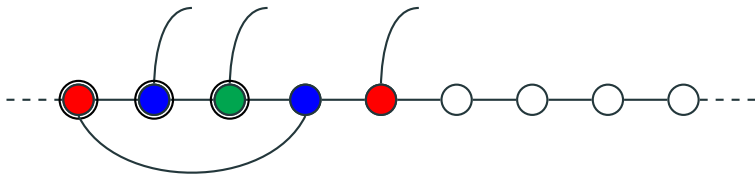
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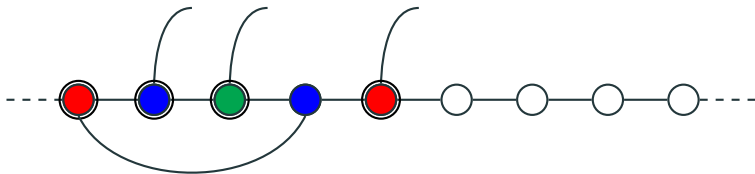
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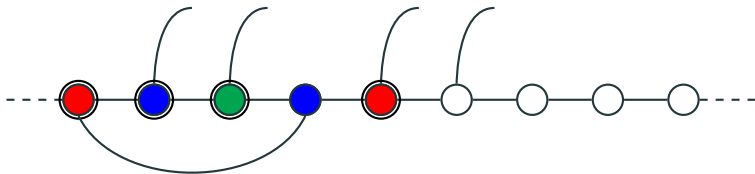
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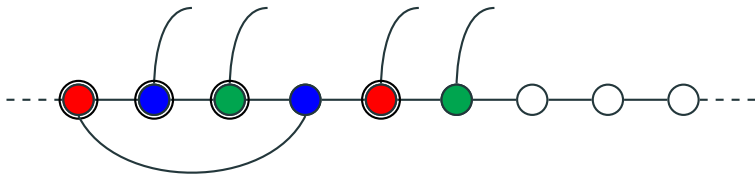
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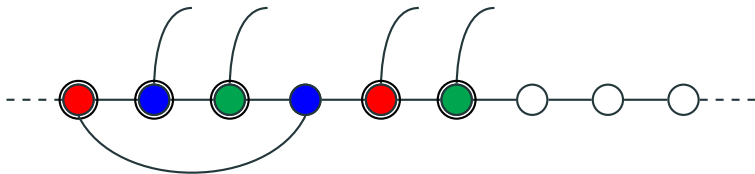
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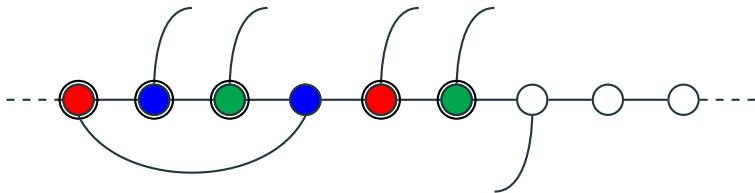
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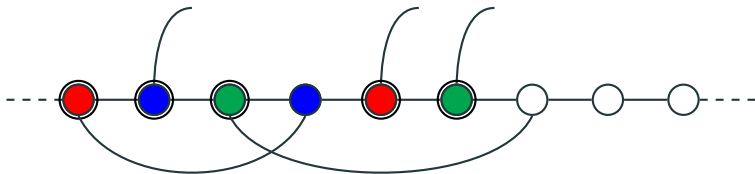
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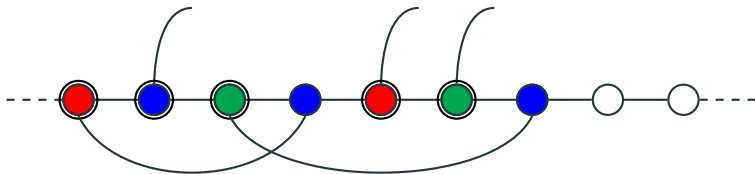
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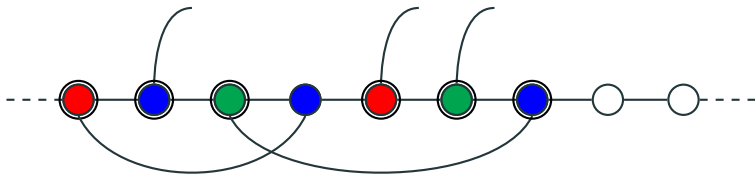
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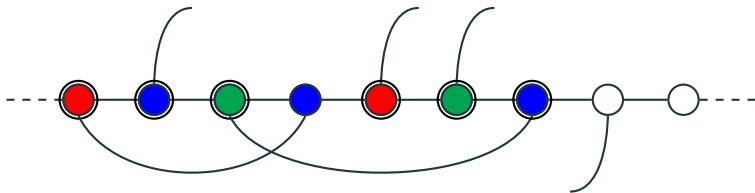
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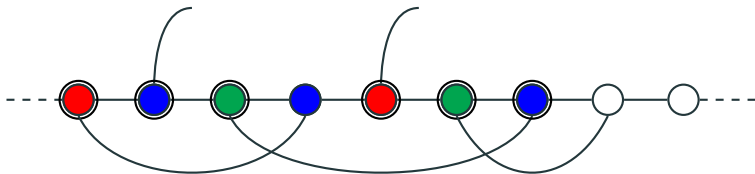
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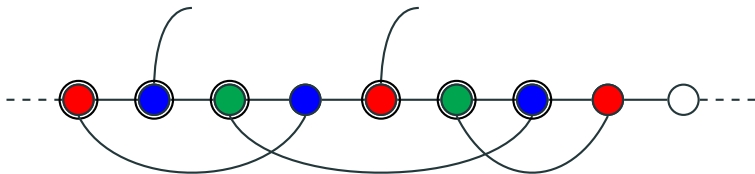
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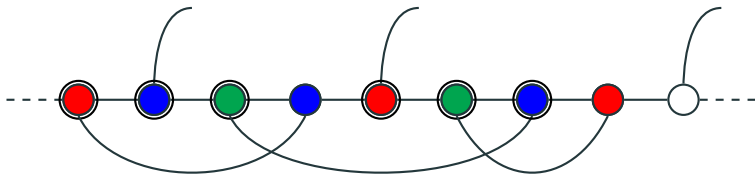
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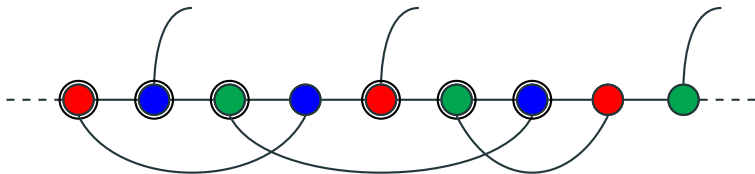
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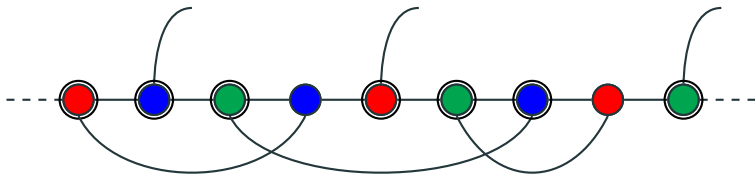
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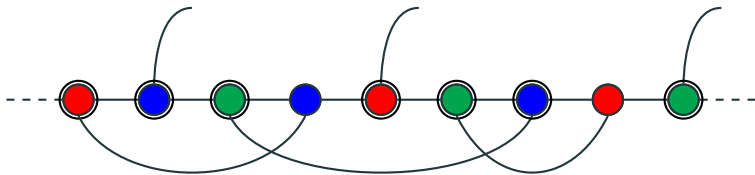
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Process vertices in the cyclic order of $\mathcal{H}_n$ and **partially reveal** $\mathcal{M}_n$ **sequentially**.



Only **bad vertices**—those with forward edges or a bad backward edge—are added to the Sudoku set.

The “greedy” Sudoku set

If S is the greedy Sudoku set,

$$\begin{aligned} |S| &= \# \text{forward edges} + \# \text{bad backward edges} \\ &= \frac{n}{2} + \# \text{bad backward edges} \end{aligned}$$

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If $X_c(i) = (1 + o(1))\frac{X(i)}{3}$ for all $c \in \{r, b, g\}$ and all i

$$\mathbb{P}(\text{vertex } i \text{ is bad} \mid i \text{ has a backward edge}) = (1 + o(1))\frac{1}{3}$$

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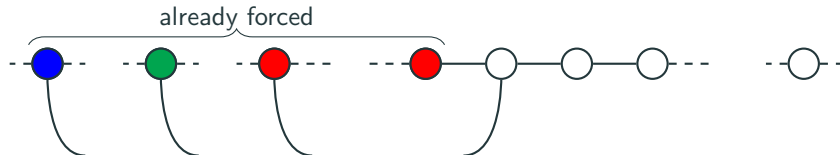
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$$\# \text{bad backward edges} = (1 + o(1)) \frac{n}{2} \cdot \frac{1}{3} \text{ a.a.s.} \implies |S| = (1 + o(1)) \frac{2n}{3}$$

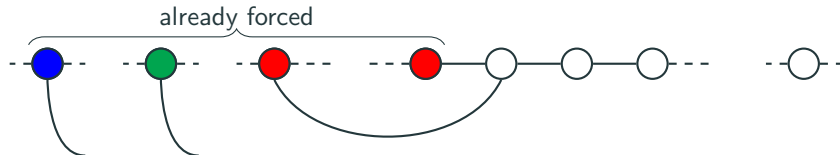
The “greedier” Sudoku set

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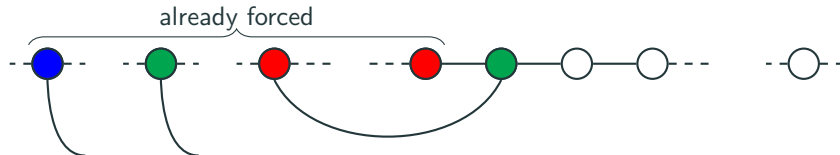
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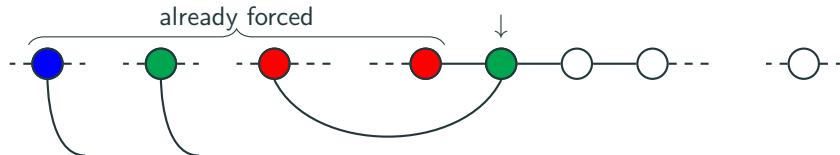
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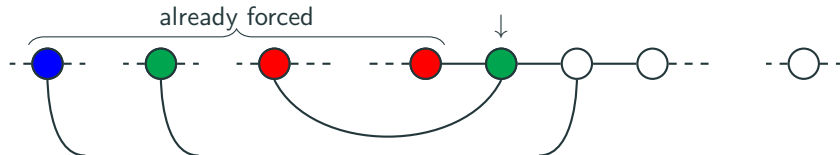
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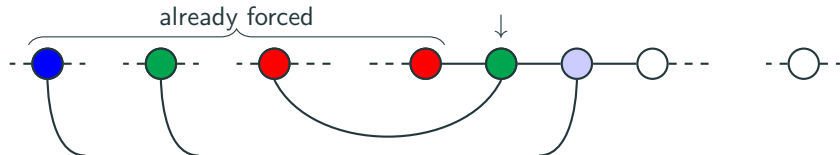
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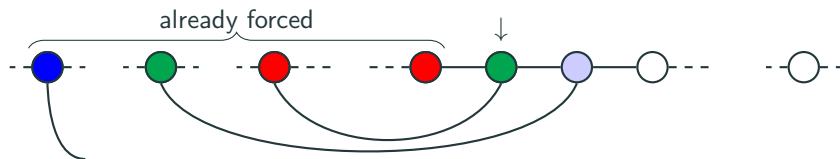
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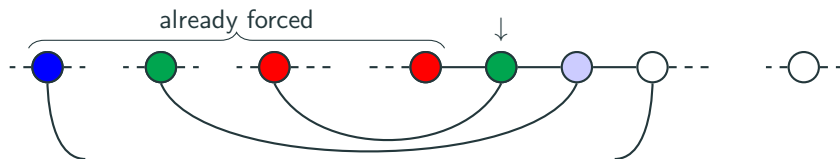
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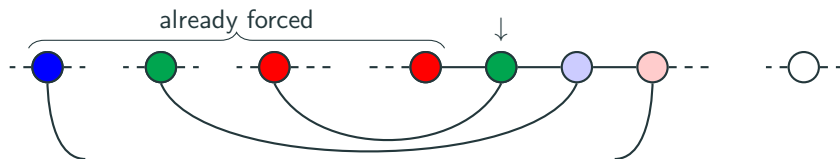
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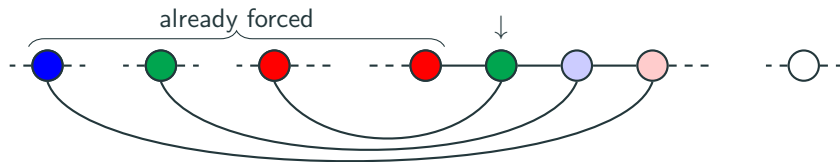
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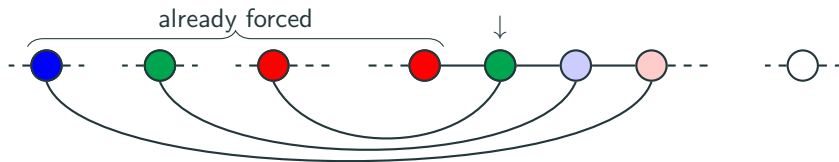
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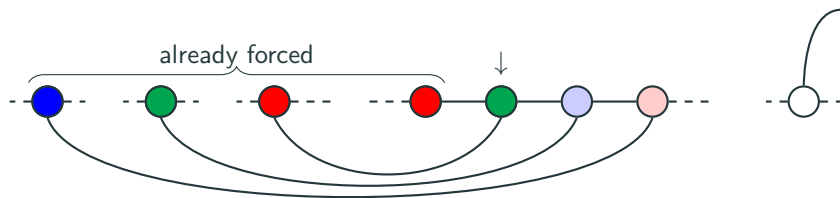
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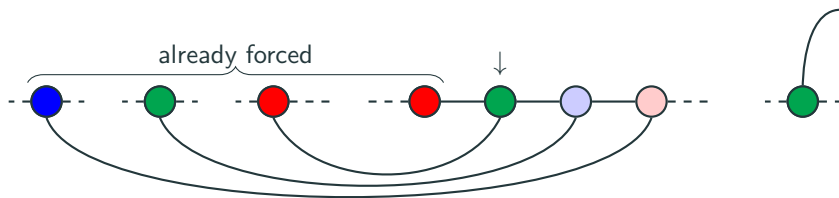
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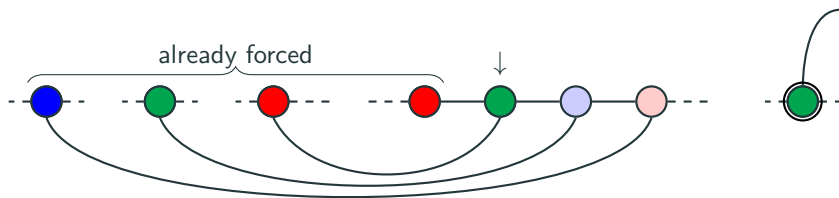
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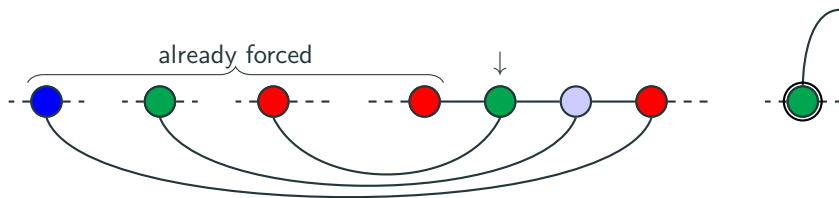
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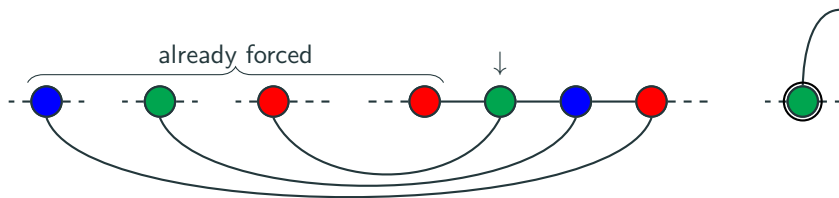
If at first you don't succeed...



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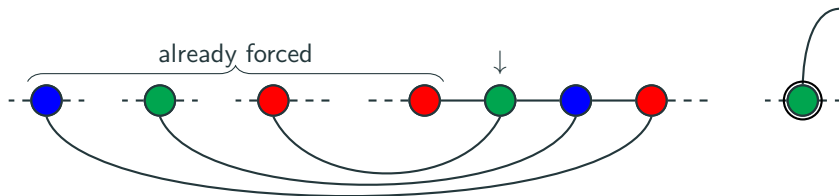
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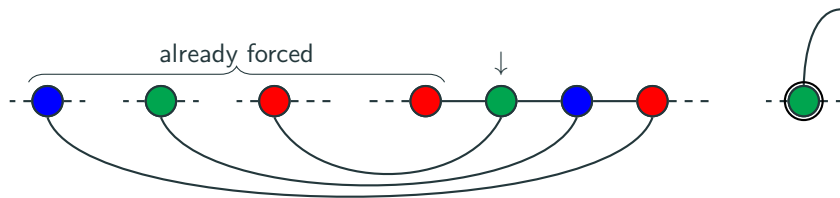


Carry on like this until a **forward edge** or **bad backward edge**—putting the bad vertex in S causes a cascade of forces from right to left.

We now add **every other** bad vertex, which gives $|S| = (1 + o(1))\frac{n}{3}$.

The “greedier” Sudoku set

If at first you don't succeed...



Still crucial for the analysis that $X_c(i) = (1 + o(1)) \frac{X(i)}{3}$ for all $c \in \{r, b, g\}$ and all i !

Analysis

We show

$$\frac{1}{\omega} \mathbb{E} [X_c(i + \omega) - X_c(i) \mid \mathcal{H}_i] = (1 + o(1)) F\left(\frac{X(i)}{n}, \frac{i}{n}\right)$$

where $\omega = o(n)$ and $F(x, t) = \frac{1}{3} \left(1 - \frac{2x}{1-t}\right)$ is a “nice” function.

Approximate the algorithm by a Markov chain which mixes in time $o(\omega)$.

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Let $t = i/n$, $x_c(t) = X_c(i)/n$, and $x(t) = X(i)/n$.

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Diff. eq. method $\implies X_c(i) = (1 + o(1)) n x_c(t) = (1 + o(1)) \frac{n x(t)}{3}.$

Conclusion

Summary of Results

We show that

$$s(\mathcal{G}_{n,3}) \leq (1 + o(1))\frac{n}{3}$$

a.a.s., and have conjectured that

$$s(\mathcal{G}_{n,3}) = (1 + o(1))\frac{n}{4}$$

a.a.s., which is asymptotically the smallest possible for cubic graphs with chromatic number 3 on n vertices.

Future directions

- Conjecture
- Planted models?
- $\mathcal{G}_{n,d}$ for $d \geq 4$?

Thank You!
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